

Stuart Khaw

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Education

National University of Singapore (NUS), *Bachelor of Computing in Computer Science*. 2025 – May 2029

GPA: 4.0/5.0 (*Second Class Honours, Upper Division*).

Relevant Coursework: CS1101S (*Programming Methodology I*), CS2030S (*Programming Methodology II*), CS2040S (*Data Structures & Algorithms*).

Academic Awards: NUS Merit Scholarship

Experience

Private Mathematics Tutor | December 2024 – Present | *Freelance*

- Conducted 300+ hours of one-on-one mathematics tutoring for secondary students across syllabus topics.
- Improved students' confidence/academic performance through structured explanations and targeted feedback.
- Identified recurring gaps in traditional tutoring workflows, which inspired involvement in educational technology initiatives like PrecisionMaths.

Projects

PrecisionMaths | *Python, PostgreSQL, TypeScript* April 2026 – Present

- Developed utilities and backend tooling for automated generation/validation of mathematics question banks.
- Contributed to scalable worksheet generation pipelines supporting configurable problem types/difficulty levels.
- Applied mathematical logic and algorithmic techniques to ensure correctness, consistency and diversity of generated outputs.
- Assisted with ingestion and seeding workflows for structured educational content into a PostgreSQL database.
- Collaborated across technical and non-technical areas in a fast-paced startup environment, contributing to both engineering and product execution.

SmokeSmart | *JavaScript, HTML, CSS* Jan 2026 – Feb 2026

- Developed a prototype web-based dashboard using agentic AI concepts to support urban planning and public health decision-making.
- Built interactive software features visualizing environmental and demographic data.
- Designed recommendation workflows to identify optimal locations for new designated smoking areas and evaluate existing infrastructure placement.
- Collaborated in a team environment to deliver a functional proof-of-concept prototype demonstrating AI-assisted urban planning systems.

Technical Skills

Languages: Java, Python, C/C++, SQL (Postgres), JavaScript, HTML/CSS

Developer Tools: Git, VS Code, IntelliJ

Activity and Awards

RC4 Pitching Competition 2025 | Winning Team | Pitched PrecisionMaths, an educational technology initiative focused on scalable and automated mathematics learning workflows, to a panel of judges in a competitive innovation showcase.

SMU Hack for Cities 2026 | Finalist | Developed SmokeSmart, an AI-assisted urban planning prototype focused on optimizing designated smoking infrastructure using environmental and demographic data.

CS1101S Sumobot | 1st Runner-Up | Designed and programmed an autonomous robot for competitive real-time navigation and strategy challenges.